# Variation

Variation: Differences between individuals of the same species.

Continuous variation: Results in a range of phenotypes between two extremes.

Example: body length / Body mass

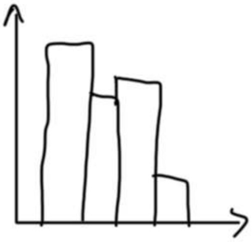
Characteristics of continuous variation:

1- Controlled by both genes and environment

Genes: * Sexual reproduction
* Meiosis: Production of genetically different gametes.
* Mutation: Random change in sequence of DNA bases
→ Changes protein

2- There is a range of phenotypes: * Intermediate values

3- Controlled by large numbers of alleles, represented on histograms:



Discontinuous variation: Results in a limited number of phenotypes with no intermediates.

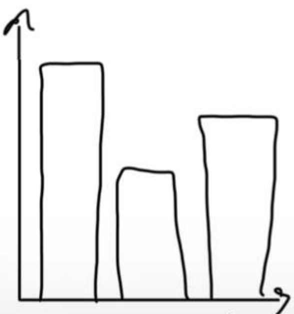
Example: ABO blood groups, seed shape and color in peas

Characteristics of discontinuous variation:

1- Only controlled by genes.

2- There are distinct groups of phenotypes (Limited)

3- Controlled by a few numbers of alleles, represented by bar charts:



Mutation: Genetic change that results in new alleles formation

Gene mutation: Random change in the base sequence of DNA

→ Ionising radiation and some chemicals increase the rate of mutation

Adaptive feature: Inherited feature that helps an organism to survive and reproduce in its environment

Adaptations of Xerophytes:

1- Stem is green, swollen, fleshy

2- Roots are long and deep to absorb water deep into the ground. Or shallow and wide to absorb water immediatly after it rains.

3- leaves: * Small surface area / Needle shaped

* Thick cuticle / large distance of diffusion.

* Stomata is at lower epidermis of the leaf

* Stomata closes during the day

Advantage: No transpiration → less water loss

Disadvantage: Less photothensis because it stops $CO_2$

* Rolled leaves → high humidty → small concentration gradient

Adaptations of hydrophytes:

Leaves:

* large air spaces between spongy cells, so they float to absorb more sunlight for more photosynthesis

* Stomata on upperepidermis so that gas exchange or diffusion of gases in air.

* Thin cuticle layer, as there is no need to conserve water
* large broad leaf, so that plant can float to trap more light.

No/little stem and roots because plants are supported by water

# Natural Selection
↳ Causes evolution

Large population of an organism → There is a variation up to mutatio → Under selection pressure: Heavy rain, draught, predators, High temperature, Shortage of food.

Adaptation: The process resulting from natural selection, by which Populations become more suited to their environment over many generations.

Some individuals can survive as they can adapt, because they have adventageuous allele, which helps them adapt. Only adapted individuals can survive and reproduce, passing the adventageous allele to their offspring. Overtime, this allele passes from generation to generation, which will change allele frequency → Evolution

Most of the population died because they cannot adapt

# Natural Selection

Large populations of bacteria → There is variation up to mutation → Under selection pressure: Antibiotics

Some bacteria survive because they have resistant allele. Only resistence bacteria survive and reproduce, passing resistant allele to their offspring. Overtime, resistant allele passes from Generation to generation. So, change in allele frequency. → Evolution

Most bacteria die because they cannot adapt

# Artificial selection
## ↳ Selective breeding

Selecting individuals with desired characteristics, breed these individuals together. From the offspring, choose the individuals with desired characteristics and breed them. Repeat the process many times, select and breed.

## Differences between natural and artificial selection:

**Natural:**

* Occurs naturally
* Results in adaptations that enhance survival and reproduction in a specific environment
* Can lead to the gradual change and diversification of species over time

**Artificial:**

* Occurs with human intervention and selective breeding
* Focuses on traits that are advantageous or desirable from a human prespective
* Can result in rapid changes in traits within a population over a relatively short period